EXP - 6 BIDIRECTIONAL RNN VS FEED FORWARD NEURAL FOR TIME-SERIES PREDICTION

AIM:

To implement a Bidirectional Recurrent Neural network (BiRNN) and a feed forward Neural network (FFNN) for time-series prediction and compare their performance using Mean Squared Error (MSE).

SOFTWARE REQUIREMENT:

* Python 3.x
* Pytorch
* Numpy
* Pandas
* Matplotlib
* Scikit-learn
* Airline Passenger Dataset

ALGORITHM:

* import the requirement libraries
* Load the airline passenger dataset
* Normalize the passenger data
* Create sequences from the time-series data
* Divide the dataset into training and testing sets
* Build a BiRNN model and FFNN model
* Train both models using the Adam optimizer
* Predict passenger numbers
* Calculate MSE for both models
* plot and compare the predictions

PROGRAM:

```
import numpy as np
import pandas as pd
import torch
import torch.nn as nn
```

```
import matplotlib.pyplot as plt

from sklearn.preprocessing import MinMaxScaler
from sklearn.metrics import mean_squared_error

df = pd.read_csv(url, usecols=[1])
data = df.values.astype("float32")

scaler = MinMaxScaler
data = scaler.fit_transform(data)

seq_len = 10
x = np.array([ data[i : i+seq_len]
    for i in range(len(data)-seq_len)])

x_train = torch.tensor(x[:split])
y_train = torch.tensor(y[:split])
x_test = torch.tensor(x[split:])
y_test = y[split:]

class BiRNN train(model, x, y):
    optimizer = torch.optim.Adam(
        model.parameters(), lr=0.01
    )
    loss_fn = nn.MSELoss()

    for epoch in range(100):
        model.train()
        optimizer.zero_grad()

        output = model(x)
        loss = loss_fn(output, y)

        loss.backward()
        optimer.step()

    return model

birnn = train(BiRNN(), x_train, y_train)
ffnn = train(FFNN(), x_train, y_train)
```

Expected Output:

BiRNN MSE: ...

FFNN MSE: ...

```
birnn.eval()
ffnn.eval()
with torch.no_grad():
    pred1 = birnn(x_test).numpy()
    pred2 = ffnn(x_test).squeeze(-1)).numpy()

pred1 = scaler.inverse_transform(pred1)
pred2 = scaler.inverse_transform(pred2)
actual = scaler.inverse_transform(y_test)

print("BiRNN MSE:", mean_squared_error(actual, pred1))
print("FFNN MSE:", mean_squared_error(actual, pred2))

plt.plot(actual, label="Actual")
plt.plot(pred1, label="BiRNN")
plt.plot(pred2, label="FFNN")

plt.xlabel("Time Step")
plt.ylabel("passengers")
plt.title("BiRNN Vs FFNN")
plt.legend()
plt.show()
```

RESULT:

The BiRNN and FFNN models were implemented for airline passenger forecasting. Their prediction performance was compared using Mean Squared Error.